

Section 1 — Academic Papers

Towards Automated Cross-domain Exploratory Data Analysis through Large Language Models

Jun-Peng Zhu et al., 2025, PVLDB.

Problem: Data analysts face challenges in crafting complex SQL queries and generating suitable visualizations across diverse database schemas.

Approach: Introduces TiInsight with hierarchical data context (HDC) via LLMs, question clarification, text-to-SQL (TiSQL), and TiChart visualization; deployed in production with GUI.

Relevance: Directly supports automated SQL-based EDA, profiling, visualization, and narrative insights for databases.

arXiv: <https://arxiv.org/abs/2412.07214> (verified). [verified]

LLM/Agent-as-Data-Analyst: A Survey

Zirui Tang et al., 2025, arXiv.

Problem: Traditional data analysis lacks natural language interfaces, semantic understanding, and autonomous orchestration.

Approach: Surveys LLM/agent techniques for structured, semi-structured, unstructured, and heterogeneous data; outlines semantic-aware design, tool-augmented workflows.

Relevance: Comprehensive overview of LLM agents for analysis, narrative generation, and EDA-like tasks.

arXiv: <https://arxiv.org/abs/2509.23988> (verified). [verified]

An LLM-Based Approach for Insight Generation in Data Analysis

A.S. Pérez et al., 2025, arXiv.

Problem: Generating actionable textual insights from multi-table databases.

Approach: Uses LLMs to produce concise, pattern-reflecting text insights from tabular data.

Relevance: Focuses on narrative findings and recommendations from profiled data.

arXiv: <https://arxiv.org/abs/2503.11664> (verified). [verified]

InsightLens: Augmenting LLM-Powered Data Analysis with Interactive Insight Management

L. Weng et al., 2024, arXiv.

Problem: Managing and navigating insights in conversational LLM data analysis

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Approach: Augments LLMs with interactive insight management for better exploration.

Relevance: Enhances LLM-driven EDA with interactive narrative and visualization.

arXiv: <https://arxiv.org/abs/2404.01644> (verified). [verified]

LIDA: A Tool for Automatic Generation of Grammar-Agnostic Visualizations and Infographics

Victor Dibia, 2023, arXiv (ACL Demo).

Problem: Automating effective visualizations from data semantics.

Approach: Multi-module system (summarizer, goal explorer, visgenerator) using LLMs for code generation/repair across grammars.

Relevance: Core for auto-visualization and narrative-infographic reports.

arXiv: <https://arxiv.org/abs/2303.02927> (verified). [verified]

AdaVis: Adaptive and Explainable Visualization Recommendation

S. Zhang et al., 2023, arXiv.

Problem: Providing explainable chart recommendations for tabular data.

Approach: Logical reasoning-based adaptive recommendation.

Relevance: Supports intelligent visual report generation.

arXiv: <https://arxiv.org/abs/2310.11742> (verified). [verified]

A Data-centric AI Framework for Automating Exploratory Data Analysis and Data Quality Tasks

Authors (ACM JDIQ), 2023.

Problem: Pain points in manual EDA and data quality for AI.

Approach: Framework with novel algorithms for automation based on user interviews.

Relevance: Directly addresses profiling, quality issues, and automated EDA.

DOI: <https://doi.org/10.1145/3603709> (verified). [verified]

Additional relevant works include surveys on visualization recommendation and data quality tools (e.g., 2022 surveys on autoEDA R packages and DQ measurement).

Section 2 — GitHub Repositories

Data profiling libraries

- **ydata-profiling** (<https://github.com/ydataai/ydata-profiling> or successor **fg-data-profiling**) — ~10k+ stars (observed ~2025/2026). One-line comprehensive reports with stats, visuals, correlations, missing values, and quality alerts for Pandas/Spark. Active. Excels at detailed profiling and HTML reports; lacks deep LLM narrative or direct DB connections in core. [verified]
- **Sweetviz** — High stars. Generates beautiful target/comparison visualizations and associations for EDA. Active. Strong visuals and comparisons; limited quality rules and no LLM. [verified]
- **DataPrep** — EDA and preparation with visualizations. Active-ish. Good low-code prep; less focus on quality anomalies. [verified]
- **capitalone/DataProfiler** — Schema, stats, entity (PII) detection. Active. Strong for sensitive data; narrower than full EDA. [verified]

LLM-powered data agents

- **PandasAI** — 20k+ stars. Natural language queries on dataframes/databases with LLMs/RAG. Very active. Excellent conversational analysis; can hallucinate on stats, security risks with code exec. [verified]
- **Vanna** — Text-to-SQL with RAG/agentic retrieval (archived ~2026 but influential). Accurate SQL for DBs. Strong training on schemas; focuses on queries over full profiling. [verified]
- **LIDA** — LLM-driven visualization and infographic generation. Active. Grammar-agnostic charts + goals; integrates well with narrative. [verified]

Auto-visualization / narrative reporting

- **LIDA** (as above).
- Related forks/extensions for Streamlit integration.

Data quality / anomaly detection

- Tools like Great Expectations (implied in ecosystem) pair with profilers for validation.

Full-stack EDA report generators

- **D-Tale** (man-group/dtale) — Interactive web explorer for dataframes. Strong interactivity.

- **AutoViz** (AutoViML/AutoViz) — Automated visualizations.

Closest matches combine profiling + LLM (e.g., PandasAI + ydata-profiling pipelines).

Many are active but lack seamless DB + LLM narrative + quality + visuals in one tool.

[verified where linked]

Section 3 — Commercial / Closed-Source Tools

- **Julius AI** (julius.ai): AI data analyst for Excel/CSV/Google Sheets; natural language analysis, charts, models, summaries. Strong overlap in narrative/reports; file-focused, not heavy DB. Pricing: Subscription tiers.
- **Akkio**: No-code AI for data prep, predictive modeling, chat with data, insights. Good for profiling/predictions; agency-focused. Enterprise pricing.
- **Hex**: AI analytics platform with notebooks (SQL/Python), Magic AI for analysis, apps/dashboards. Excellent collaborative EDA; connects data sources. Pricing: Usage-based.
- **Tableau Pulse** (Salesforce): AI-powered metric insights, personalized digests, natural language explanations in workflows (Slack/Teams). Strong narrative/proactive insights; metric-centric. Part of Tableau licensing.
- **Mode Analytics**: Modern BI with SQL/Python/notebooks, AI features. Overlaps in reporting/EDA. Enterprise.

Others like Athenic (query-focused) exist in the space.

Section 4 — Gap Analysis

1. Well-covered: Classical profiling/stats (ydata-profiling, Sweetviz), basic auto-viz (LIDA, AutoViz), text-to-SQL/natural language querying (PandasAI, Vanna), interactive exploration (D-Tale, Hex).
2. Weakly covered: End-to-end seamless DB + CSV connection with automated quality (drift, schema, cardinality, outliers) + LLM "Senior Data Scientist" narrative recommendations + integrated visual reports with minimal hallucination. Full autonomy across heterogeneous sources with security.

3. Strongest differentiator: Reliable, auditable LLM agent as "Senior Data Scientist" that grounds narratives in verified stats/quality checks, handles schema drift/security for real DBs, and produces publication-ready reports with citations to underlying computations.
4. Risks: LLM hallucination on statistics (common in PandasAI-style tools), scale for large DBs, security of executing generated code/queries on production DBs, cross-domain generalization, maintaining freshness against drift.

Section 5 — Suggested Reading Order

1. **TilInsight paper** — Most production-relevant LLM + SQL + viz EDA system.
2. **LLM/Agent-as-Data-Analyst Survey** — Broad landscape of LLM approaches.
3. **ydata-profiling GitHub/docs** — Foundation for classical profiling/quality.
4. **LIDA GitHub/paper** — Best for auto-viz + infographics.
5. **PandasAI GitHub** — Leading LLM agent example (and its limitations).